Nazim Belabbaci

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Summary

I am Research Engineer and PhD Candidate specializing in IoT, Biosensors, and AI for healthcare. Experienced in developing end-to-end systems, from sensor design and embedded systems to signal processing and machine learning deployment. I have a growing publication record, hands-on prototyping experience, and clinical research. I am interested in roles where I can contribute to meaningful, data-driven work at the intersection of AI and human health.

Research & Academic Experience

Graduate Research Assistant, *CUBICS Lab, University of Massachusetts Lowell*, Lowell, MA| 2022–Present

◊ *Wearable Design*: Designed a microcontroller-driven wearable device for non-invasive hydration tracking using optical sensors;
patent pending co-inventor of "Methods for Body Hydration Estimation Using Wearable Sensors".

◊ Leading a controlled exercise-based hydration study for PhD dissertation; drafted and secured IRB approval, managed data collection pipelines, and ground-truth validation using saliva osmolarity.

◊ *Generative AI for Biosensing*: developed a diffusion model to generate synthetic FTIR spectra. This data augmentation strategy boosted cancer detection classifier accuracy by 4% and specificity by over 12% while maintaining sensitivity. Code and pre-trained models open-sourced on my GitHub.

◊ Developed Transformer-based architectures for real-time classification of fatigue and mood using physiological time-series data, achieving a 3-10% performance improvement over benchmarks.

PhD Research Intern, *Beth Israel Deaconess Medical Center (BIDMC)*, Boston, MA| 2025–Present

◊ Investigated splicing events in 5 viral models by building a HPC pipeline (FASTQ, STAR, rMATS).

◊ Complemented this with custom Python tooling to automate statistical filtering and downstream functional analysis, effectively identifying key isoform switching events.

Graduate Teaching Assistant, *University of Massachusetts Lowell*, Lowell, MA| 2023–Present

◊ COMP 2030 Assembly Language & COMP 4200 Artificial Intelligence: Led labs and recitations for 100+ students; graded assignments/exams; held office hours and created supplemental materials.

Graduate Research Assistant, *Takaku Lab, University of North Dakota,* Grand Forks, ND| 2022

◊ Built interpretable ML models to identify breast-cancer-associated transcription factor binding.

◊ Collaborated to build MotifXplorer, a clinician-friendly web platform for visualizing regulatory sequence predictions and model outputs.

Prior Technical Experience

Co-Founder & Embedded Systems Engineer, Deadline Technologies, Bouira, DZ| 2020–2021

◊ Founded a hardware startup specializing in wireless embedded boards for academic and industrial use.

◊ Managed the full product lifecycle: from PCB schematic design and firmware development to manufacturing and sales.

Embedded Software Engineer, EURL Microtechnologies Lab, Boumerdès, DZ | 2019–2020

◊ Developed firmware and software for laboratory-grade electrical measurement instruments used in national educational institutions.

Education

PhD, Computer Science, University of Massachusetts Lowell, Lowell, MA | 2023–Present
(*Thesis: "Hardware-Enabled Predictive Framework: Advancing Health Monitoring and Disease Diagnosis with Conditional Diffusion Models and Deep Learning"*) (GPA 3.98)
Master, Computer Science, University of Massachusetts Lowell, Lowell, MA | 2023
(Thesis: "A Prototype Smartwatch for Real-Time Monitoring of Hydration Status") (GPA 3.98)
Master of Computer Engineering, National Institute of E&E Engineering, Boumerdès, DZ | 2019
(*Thesis: "Low-cost micro-controller-based Phasor Measurement Unit*") (GPA 3.7)
Bachelor in E&E Engineering, National Institute of E&E Engineering, Boumerdès, DZ | 2017
(*Thesis: "Eco-Friendly Voice-Controlled Smart Home"*) (GPA 3.7)

Awards & Honors

Finalist of the $200K Challenge**,** an annual start-up competition hosted by **M2D2** |2024**.**
LaTorre Family Scholarship Endowment fund award winner, UML endowed |2023.
Finalist of the UND Middleton School of Entrepreneurship Business Plan Competition |2022
◊ Developed and pitched a 30-page AI-based COVID-19 early-detection business plan using wearable sensor data. Conducted market analysis and financial modeling for the proposed venture.
Winner of Leapfrog International Hackathon of Algiers |2018
◊ Led a 48-hour team sprint to develop "eHealth" an IoT medical monitoring prototype and presented it to a panel of industry experts.

Civic Engagement

Chair, Student's Initiative, *AAASTE*, Pembroke Pines, FL |2022-2024
◊ Led a team of 20 to deliver online STEM webinars to 200+ students.

President & Head Project Manager, *Wameedh Scientific Club*, Boumerdès, DZ |2016-2019
◊ Secured funding and ran hands-on workshops and online classes for 100+ student.